

How Far Can General-Purpose LLMs Go in Tigrinya? A Lexicon-Grounded Evaluation with Trivial Baselines

Hailay Kidu Teklehaymanot

AFFILIATION_REQUIRED

EMAIL_REQUIRED & Wolfgang Nejdl

AFFILIATION_REQUIRED

EMAIL_REQUIRED & Gebrearegawi Gebremariam

AFFILIATION_REQUIRED

EMAIL_REQUIRED

October 4, 2026

Abstract

We evaluate eight general-purpose large language models on three lexicon-grounded Tigrinya tasks — part-of-speech tagging, morphosyntactic labelling, and English→Tigrinya translation — using a 5,783-row annotated bilingual lexicon. Our central methodological finding is that such evaluations are easy to over-read: with a permissive matching rule, models appear to reach 80–89% accuracy, but the same outputs score 0–82% under a first-token rule, and a constant-answer baseline reaches 53.7% while an English-only tagger that never sees the Tigrinya reaches 60.4%. Measured against those baselines, three of eight models demonstrate above-trivial POS performance and two exceed the English-only baseline on morphosyntactic labelling, by under three points. On translation all eight models score at or below chrF 0.81, while NLLB-200-3.3B, which supports Tigrinya, reaches chrF 16.05 on identical items with an identical scorer. We argue that low-resource evaluations without trivial baselines are uninterpretable, and release the lexicon, code, model outputs, and job logs. Because the procedure we apply here — report every matching rule, measure against baselines that use none of the target language, and quantify prompt echo — requires no resources beyond those an evaluation already has, it transfers directly to other low-resource languages, where the same scoring choices determine whether a reported number reflects competence or an artefact of measurement.

1 Introduction

Tigrinya is a Semitic language of Eritrea and northern Ethiopia with approximately 9–10 million speakers and very little machine-readable linguistic infrastructure. This scarcity makes it a natural test of whether general-purpose LLMs generalise beyond high-resource languages, and a difficult one to measure: without established benchmarks it is easy to produce numbers that look like competence.

This paper makes three contributions.

1. **A lexicon-grounded benchmark** of 5,783 English–Tigrinya rows (4,331 English head-words, 3,767 Tigrinya forms), each carrying a part-of-speech label, derived from a digitised bilingual dictionary (Union of International Democrats, 2015).
2. **An evaluation of eight LLMs** across POS tagging, morphosyntactic labelling and translation, reported under three matching rules and against two trivial baselines.

3. **A methodological result:** on a lexicon-scale task, the choice of matching rule moves reported accuracy by up to 26 points, and two baselines that use no Tigrinya at all reach 53.7% and 60.4%. Papers in this setting that report neither are not interpretable.

The third contribution is the reason for this paper. An earlier version of this work reported results computed with a permissive matching rule and no baselines, and drew an architectural conclusion that does not survive either correction. The authors have requested its retraction; Section 7 states the relationship between the two.

2 Related work

2.1 Tigrinya and Ge’ez-script NLP

Tigrinya remains severely underrepresented in NLP: a recent survey of over 50 studies from 2011 to 2025 finds a field whose progress is driven almost entirely by discrete resource-creation milestones rather than by modelling advances (Gaim and Park, 2026), a pattern echoed in broader surveys of Ethiopian-language NLP (Tonja et al., 2023). The resources that exist are small and task-specific. Öktem et al. (2020) built a domain-specific Tigrinya→English NMT system using transfer learning from other Ge’ez-script languages, reporting a 1.3 BLEU improvement over a neural baseline; Tela et al. (2020) showed that an English XLNet fine-tuned on 10k examples reaches 78.88 F1 on Tigrinya sentiment, outperforming mBERT by 7 points; TiGQA contributed 2,685 expert-annotated question–answer pairs from 537 context paragraphs (Teklehaymanot et al., 2024); and VEXMLM demonstrated that vocabulary extension with 30k Ge’ez-script subwords substantially improves XLM-R on Amharic and Tigrinya, attributing the baseline weakness to high out-of-vocabulary rates under Latin-centric tokenizers (Teklehaymanot et al., 2026b). Tigrinya also appears as one slice of multilingual benchmarks — MasakhaPOS, the largest POS dataset for African languages, covering 20 typologically diverse languages under UD guidelines (Dione et al., 2023), and Belebele, a parallel multiple-choice reading-comprehension set in 122 language variants (Bandarkar et al., 2024). Our lexicon complements these: it is not running text, and it tests the dictionary-headword level that sits beneath all of them. The structural reason such resources are scarce at all is the one Joshi et al. (2020) documented — representation in NLP venues tracks resource availability, calling into question the “language agnostic” status of current systems.

2.2 LLM evaluation for low-resource languages

NLLB-200 added Tigrinya (`tir_Ethi`, Ge’ez script, Afro-Asiatic/Semitic) to a 200-language translation model (NLLB Team et al., 2022), evaluated on FLORES, whose professionally translated, multilingually aligned design set the standard for low-resource MT benchmarking (Goyal et al., 2022). Against that backdrop, several studies have established that general-purpose LLMs lag purpose-built MT on exactly the languages we study. Robinson et al. (2023) evaluated ChatGPT on 204 FLORES-200 varieties and found NLLB superior on 169 of 201 languages (84.1%), by an average of 11.9 chrF, with the gap widest for African languages and the Ge’ez script specifically; they further report that five-shot prompting improved on zero-shot by only 0.88 chrF on average across 203 directions, which is consistent with our own few-shot probe moving gemma-7b from chrF 0.16 to 0.66 with exact match unchanged. Hendy et al. (2023) reach the same conclusion for GPT models across eighteen directions — competitive for high-resource, limited for low-resource — and Zhu et al. (2023) find GPT-4 beating NLLB in 40.91% of directions overall while still trailing badly on the long tail. Adebara et al. (2025) extend this to a continent-scale African benchmark and attribute the residual disparities to data inequities rather than model scale. Our Section 5.3 result — three NLLB checkpoints at chrF 14.1–16.1

against eight LLMs at ≤ 0.81 — is this finding at lexicon granularity, where the LLM floor is not merely lower but absolute.

A separate strand asks whether LLM evaluations in these settings are conducted carefully enough. LArABench benchmarked 33 Arabic tasks across 61 datasets and explicitly computed a random baseline “to determine if the LLMs predictions are not merely the result of chance” (Abdelali et al., 2023). We regard that step as mandatory rather than optional, and Section 5 shows why.

2.3 Trivial baselines and spurious cues

The methodological core of this paper has a direct precedent in natural language inference. Poliak et al. (2018) showed that a hypothesis-only model — one that never sees the premise, a degenerate solution by construction — significantly outperforms a majority-class baseline on a number of NLI datasets; Gururangan et al. (2018) showed the same artefacts let a simple text classifier label the hypothesis alone correctly in about 67% of SNLI and 53% of MultiNLI, concluding that model success “has been overestimated”. McCoy et al. (2019) demonstrated that strong test-set scores can rest on fallible syntactic heuristics that collapse on controlled sets, and Niven and Kao (2019) found BERT’s 77% on argument reasoning comprehension to be entirely accounted for by spurious statistical cues. Our English-only tagger is a hypothesis-only baseline in this sense: the prompt format `Phrase: {Tigrinya} ({English})` leaves the part of speech largely recoverable from the gloss, and the baseline reaching 60.38 is the lexicon-task analogue of the 67% SNLI result.

For POS tagging specifically, Kann et al. (2020) make the closest argument to ours: weakly supervised taggers reported to perform “almost as well as supervised ones” were evaluated on languages unlike truly low-resource ones, and on 15 genuinely low-resource languages the best model gets fewer than half the words right, with a strongest baseline macro-average of 39.11%. They also note that such methods presuppose “high-coverage and almost error-free dictionaries” — precisely the assumption our Section 3.3 annotator notes undermine, where the source dictionary contains mislabelled parts of speech and outright mistranslations. At the level of benchmark design, Reuel et al.’s assessment of 46 best practices across 24 benchmarks identifies stating random performance as a scored criterion, since without it one cannot separate capability from chance or metric design (Reuel et al., 2024).

2.4 Scoring rules, answer extraction and prompt artefacts

Our finding that the matching rule moves reported accuracy by up to 26 points on fixed outputs belongs to a growing literature on evaluation mechanics. Tam et al. (2024) show that constraining LLMs to structured output formats significantly degrades measured reasoning, and disentangle format errors from content by using an LLM as a “perfect parser” rather than a regex — the mirror image of our problem, where permissive parsing manufactures competence instead of destroying it. Kirouane and Petrocheilos (2026) document the same failure mode concretely in a low-resource setting: a scorer that scanned the whole response instead of the requested answer line produced a +29.8 pp artefact, one-directional against verbose outputs, and they conclude that “any benchmark that requests an answer format must score that format first and report how often it was absent”. Our first-token rule is that recommendation applied; our exact / first-token / set-intersection triple reports the spread it would otherwise hide.

The prompt-echo column in Section 5.2 relates to known label-copying behaviour. Zhao et al. (2021) show that few-shot LLM accuracy is destabilised by bias towards answers appearing in the prompt or common in pretraining, varying “from near chance to near state-of-the-art” with prompt format alone. When the gold distribution is 53.7% `noun` and the prompt example contains the word `noun`, a model that copies the example is scored correct by construction under

set-intersection — gemma-7b does this on 97.4% of answers. We therefore treat echo rate as a reportable diagnostic rather than a footnote.

2.5 Metrics and agreement statistics

We report chrF (Popović, 2015) because character n -gram F-score is tokenisation-independent and therefore appropriate for Ge’ez-script targets averaging 1.37 whitespace tokens, where word-level BLEU (Papineni et al., 2002) is near-undefined; we follow Post’s argument that BLEU is a parameterised family whose configurations are not comparable across papers unless the scheme is pinned down (Post, 2018), and report character-level BLEU alongside chrF for this reason. Agreement is reported as Cohen’s κ (Cohen, 1960). We note the standard caveat that κ is hard to interpret under class imbalance — the “kappa paradox” — and that absolute thresholds are not meaningful without a contextual reference (Wong et al., 2021); this is why Section 3.3 reports κ beside raw observed agreement and beside the filled-label-only variant, rather than reporting a single coefficient.

2.6 Dictionaries as evaluation resources

Digitised bilingual dictionaries are a standard bottom-up resource for languages lacking parallel corpora: Wickramasinghe and de Silva (2023) argue that for low-resource pairs it is “more feasible to move in the bottom-up direction where finer granular pairs such as dictionary datasets are developed first”, and Alnajjar et al.’s Ve’rdd addresses the complementary problem of grassroots paper dictionaries edited by many hands, which require re-evaluation rather than direct ingestion (Alnajjar et al., 2020). Our source exhibits exactly the defects that motivate that work — a direction-swapped second block, inline gender markers embedded in the target field, and annotator-flagged mistranslations — and Sections 3.1–3.3 document them rather than silently cleaning them.

2.7 Relation to the earlier version

An earlier version of this work (Teklehaymanot et al., 2026a) introduced the lexicon and the four-task framework used here. Section 7 states what it reported, which of its claims survive re-scoring, and the authors’ request for its retraction.

3 Dataset

3.1 Source and construction

The lexicon was digitised from the Swansea Tigrinya–English dictionary (Union of International Democrats, 2015), with additional entries contributed by native-speaker linguists, and carries the source dictionary’s own part-of-speech abbreviations. The digitised source holds 7,234 rows concatenated from two directions, English→Tigrinya followed by Tigrinya→English, under a single pair of column headers; in the second block the column named **English** therefore holds Ge’ez script. We detect that block by script, reorient it, and merge both directions, keeping distinct senses of one form as separate items — *estimate* → ግምት is retained as both noun and verb — which yields **5,783 gold rows** over 4,331 English headwords and 3,767 Tigrinya forms. Three evaluation set sizes follow from this and are not interchangeable: 5,783 gold rows, **5,775 items** for translation and lexical alignment after each task drops duplicate pairs and rows with an empty side, and **5,701 items** for the two tagging tasks, which were run against a further reduction representing each multi-sense entry by its first sense. Appendix A gives the row-level derivation, the provenance of the source file, and the reconstruction of the 5,701-item set from the released gold.

Dimension	κ	Observed
Part of speech	0.9076	0.9500
Gender	0.8079	0.9240
Number	0.6762	0.9280
Category agreement	0.5912	0.9000
Lexical alignment	0.8487	0.9480

Table 1: Inter-annotator agreement on the 500-item subset, blanks counted as disagreements. All five dimensions are computed over all 500 items.

3.2 What the annotations do and do not contain

The POS labels are the source dictionary’s, not independently re-annotated by us. They are coarse: **noun** accounts for 3,109 of 5,783 labelled entries (53.7%), **verb** 1,542, **adjective** 879.

Morphosyntactic features are **sparse**. Only 185 entries carry gender, number or plurality; the remaining 5,598 (96.8%) carry a value identical to the POS label. We therefore do not treat morphosyntactic labelling as a distinct linguistic dimension, and report it in Section 5.2 as what it measures: a second POS task with a different prompt.

3.3 Inter-annotator agreement

A 500-item subset of the lexicon was double-annotated for part of speech, gender, number, category agreement and lexical alignment. The records are released as `POS_english_to_tigrigna_Annotated.xlsx` — two annotator sheets, an adjudicated sheet and the annotation guide that defines each value — and `scripts/compute_iaa.py` recomputes every figure below from them.

An annotator sometimes left a cell empty where the other assigned a label. Those are not matching judgements, so we count them as disagreements and report Cohen’s κ over all 500 items, with a blank treated as its own category (Table 1).

Restricting instead to the items both annotators labelled raises every dimension — 0.9076, 1.0000, 1.0000, 0.9815, 0.9850 on 500, 420, 428, 445 and 474 items. We report the stricter figures because the difference between the two columns is largely a difference in what each annotator left blank: Annotator 1 omits 32 gender, 32 number and 46 agreement labels that Annotator 2 supplies, against 6, 4 and 3 in the other direction. Discarding those rows would hide the main source of divergence.

Three properties of this subset bound what the coefficients support.

The two columns are not independent of the adjudication. On the 25 part-of-speech items where the annotators differ, the adjudicated sheet reproduces Annotator 2 on all 25 and Annotator 1 on none; on the three gender, agreement and alignment disagreements it reproduces Annotator 1. The adjudication is assembled from the two columns rather than decided separately, so it cannot serve as an independent third judgement.

Agreement on filled labels is near-total. Where both annotators assigned a gender or number value, they never differ — 0 disagreements across 420 and 428 items. The divergence between them is entirely in coverage, not in choice of label, which is why κ falls as far as it does once blanks count.

The subset is not a random sample. Tracing each item to its source sheet shows the 500 are the alphabetical head of each category: all 9 prepositions, all 10 pronouns and all 6 interjections, then 305 adjectives and 150 nouns from the tops of those sheets. The corpus is

Label	Checkpoint	Arch.	Inst.
gemma-2b	google/gemma-2b-it (Gemma Team, Google DeepMind, 2024)	causal	yes
gemma-7b	google/gemma-7b-it (Gemma Team, Google DeepMind, 2024)	causal	yes
mistral-7b	Mistral-7B-Instruct-v0.2 (Jiang et al., 2023)	causal	yes
qwen-7b	Qwen1.5-7B-Chat (Bai et al., 2023)	causal	yes
falcon-7b	falcon-7b-instruct (Almazrouei et al., 2023)	causal	yes
mt5-small	google/mt5-small (Xue et al., 2021)	seq2seq	no
mt5-large	google/mt5-large (Xue et al., 2021)	seq2seq	no
byt5	google/byt5-small (Xue et al., 2022)	seq2seq	no

Table 2: Evaluated models. Architecture and instruction-tuning status are perfectly confounded in this set.

54% noun and 25% verb; this subset is 61% adjective and contains **no verbs**. The coefficients therefore describe agreement on this subset and are not an estimate for the corpus. Establishing one requires a stratified re-annotation covering verbs, which we leave to future work (Section 9).

The workbook also carries the annotators’ free-text notes on 201 of the 500 items, documenting mislabelled parts of speech in the source dictionary, inflected forms given as citation forms, and outright mistranslations (*at least* → *at most*). These support the caveat in Section 3.2 that the POS labels are the source dictionary’s and were not independently verified.

4 Experimental setup

Eight models, all loaded through the Hugging Face **transformers** interface (Wolf et al., 2020) on an Ampere-class GPU node (4×A100, 32 CPU cores, 128 GB RAM) under SLURM (Table 2).

We do not compare architectures. All five causal models are instruction-tuned and all three sequence-to-sequence models are raw pretrained checkpoints; the two factors are perfectly confounded in this model set and no result here can separate them. The three base seq2seq models receive span-infilling prompts matched to their pretraining objective; they nonetheless produce almost no usable output, which we report as a property of those checkpoints on this task rather than of their architecture.

Generation is greedy with `max_new_tokens=128`, `min_new_tokens=1`. `xlm-roberta-base` (Conneau et al., 2020) was included initially but fails on all tasks — its prompts contain no `<mask>` token — and is excluded.

4.1 Scoring

Model answers are frequently verbose: gemma-7b’s median POS answer is 14 words. We report three rules.

- **exact** — the cleaned answer equals the gold label.
- **first-token** — the model’s first word is a gold label word. This is the answer the model commits to, and is our headline rule.
- **set-intersection** — any word of the answer is a gold label word. This is permissive: an answer naming several parts of speech is correct as soon as one fits. We report it only to show the gap.

4.2 Baselines

Two reference points, both of which a model must beat to demonstrate Tigrinya knowledge:

Model	exact	first-tok	set-int
mistral-7b	33.71	81.56	85.91
gemma-7b	0.00	79.39	82.44
gemma-2b	71.71	72.86	73.29
falcon-7b	26.94	40.64	44.45
qwen-7b	17.00	17.00	17.07
mt5-large	0.00	0.09	1.75
byt5	0.00	0.00	0.05
mt5-small	0.00	0.00	0.00
Majority (noun)		53.73	
English-only tagger		60.38	

Table 3: POS tagging accuracy (%) under three matching rules, with both trivial baselines.

Model	first-tok	set-int	echo
gemma-2b	64.59	78.02	1.2%
gemma-7b	63.32	89.42	97.4%
mistral-7b	54.92	81.09	3.7%
falcon-7b	39.17	76.28	19.5%
qwen-7b	34.33	54.45	0.4%
mt5-small	0.86	2.49	0.0%
mt5-large	0.11	0.74	0.0%
byt5	0.00	38.55	69.7%
Majority	52.38		
English-only tagger	61.57		

Table 4: Morphosyntactic labelling accuracy (%). The final column is how often an answer contains the prompt’s own example.

- **Majority** — always answer the most frequent gold label (**noun**).
- **English-only** — an NLTK (Loper and Bird, 2002) English POS tagger applied to the English gloss alone. The prompt format is **Phrase: {Tigrinya} ({English})**, and a dictionary headword’s part of speech is largely recoverable from the English word, so this baseline uses no Tigrinya whatsoever.

5 Results

5.1 POS tagging ($n = 5,701$)

Three models exceed both baselines: mistral-7b, gemma-7b, gemma-2b. Falcon-7b and qwen-7b fall below the majority baseline; all three seq2seq checkpoints are at or near zero.

The exact column shows how brittle these numbers are. Gemma-7B scores 0.00% exact and 79.39% first-token: it always answers in a sentence, never a bare label. A reader given only one of these columns would draw opposite conclusions.

5.2 Morphosyntactic labelling ($n = 5,701$)

Because the gold is 96.8% identical to the POS label (Section 3.2), this is effectively a second POS task.

Model	chrF	BLEU _c	BLEU _w	exact
NLLB-3.3B	16.05	14.23	0.24	16.24
NLLB-1.3B	15.01	13.29	0.23	15.00
NLLB-600M	14.11	12.59	0.25	14.42
byt5	0.81	0.01	0.00	0.05
gemma-2b	0.39	0.01	0.01	0.07
mistral-7b	0.26	0.00	0.00	0.03
mt5-small	0.20	0.04	0.02	0.05
gemma-7b	0.16	0.00	0.00	0.03
mt5-large	0.13	0.04	0.01	0.02
falcon-7b	0.13	0.00	0.00	0.03
qwen-7b	0.04	0.00	0.00	0.28

Table 5: Translation quality. All values on a 0–100 scale.

Only two models exceed the English-only baseline of 61.57, by 3.02 and 1.75 points; the remaining six fall below it, and five fall below the majority baseline of 52.38.

The final column measures how often an answer contains the prompt’s own example, **preposition**, **noun**, **singular**. Gemma-7b echoes it in 97.4% of answers; because the example contains **noun**, and **noun** is the majority gold label, set-intersection scores those echoes as correct. Its apparent 89.42 is largely an artefact of copying the prompt. ByT5’s 38.55 is the same artefact at 69.7% echo, and drops to 0.00 under first-token.

5.3 Translation, English→Tigrinya ($n = 5,775$)

Tigrinya references average 1.37 whitespace tokens, so word-level BLEU-4 is near-undefined; we report chrF and character-level BLEU (Table 5).

All eight general-purpose models are at the floor. NLLB-200, for which Tigrinya (`tir_Ethi`) is a supported target language, reaches chrF 16.05 on identical items with an identical scorer — roughly twenty times the best LLM.

Note that word-level BLEU-4 is 0.24 for NLLB-3.3B, which translates 16.24% of items exactly. **Word-level BLEU is not usable at lexicon scale**, and a paper reporting it would show a working system as scoring zero.

We verified the LLM failure is not one of output format. Few-shot prompting with eight exemplars and an explicit format constraint moved gemma-7b from chrF 0.16 to 0.66 with exact match unchanged at 0.00; extracting the Ge’ez span from verbose answers recovered the reference in 2 of 5,775 items. The median character-level similarity between model output and reference is 0.000: the models emit Ge’ez-shaped strings that are not the target words (ገህል for ብዙስ, ልልል as repetition).

5.4 Lexical alignment

We report this task for completeness but do not draw conclusions from it: 2,208 of 5,783 alignment labels (38%) are positional fallbacks (`AlignmentSource: auto`) rather than annotation. Format accuracy ranges from 0.0383 (mt5-large) to 0.5484 (gemma-2b); semantic accuracy from 0.0012 (mt5-small) to 0.8876 (mistral-7b). The two measures diverge widely for individual models, which makes the task ill-posed as currently scored.

6 Discussion

The matching rule is a reporting decision, not an implementation detail. On fixed model outputs, moving from set-intersection to first-token costs gemma-7b 3.05 points on POS and 26.10 on morphosyntax; moving to exact match costs it everything. A paper that reports one number without naming the rule has not reported a result.

Trivial baselines are not optional. Two baselines using no Tigrinya at all — a constant answer and an English tagger — reach 53.73 and 60.38 on POS and 52.38 and 61.57 on morphosyntax. Five of eight models fall below the stronger baseline on POS and six of eight on morphosyntax. Without those reference points, the same table reads as evidence of cross-lingual competence.

Prompt echo is a measurable confound. Gemma-7b reproduces the prompt’s example in 97.4% of its morphosyntax answers, and because the example contains the majority gold label, a permissive scorer rewards it. We recommend reporting echo rate wherever a prompt contains a worked example.

Purpose-built MT still dominates at the lexicon level. NLLB-200 checkpoints reach chrF 14.11–16.05 where the best general-purpose model reaches 0.81 — more than an order of magnitude on chrF. For low-resource languages, model selection dominates model size.

What we can claim about Tigrinya competence. Three of eight models show above-baseline POS performance. Two exceed the English-only baseline on morphosyntactic labelling, by 3.02 and 1.75 points; this margin is within the range a held-out re-run could move, and we therefore treat it as suggestive rather than established. None can translate into Tigrinya. Even NLLB reaches only 16.24% exact on a dictionary task, which indicates the difficulty is real rather than an artefact of our setup.

7 Relation to an earlier version of this work

An earlier version of this work was published as *LLM Probe: Evaluating LLMs for Low-Resource Languages* at LLMs4SSH @ LREC 2026, pp. 224–234 (2026.llms4ssh-1.24) (Teklehaymanot et al., 2026a). It introduced the lexicon and the four-task framework used here, reported POS accuracies of 73.0–80.0 and morphosyntactic accuracies of 70.5–77.0, assigned different model subsets to different tasks, and concluded that sequence-to-sequence models “excel in morphosyntactic analysis and translation quality”. Those figures were computed with a permissive matching rule, under which an answer counts as correct if any of its words matches a gold label word, and were reported without baselines.

Re-scored under a first-token rule and measured against a constant-answer baseline and an English-only baseline, the central claims of that paper do not hold. Only three of the eight systems exceed both baselines on part-of-speech tagging and two on morphosyntactic labelling, in each case by a few points; five and six respectively fall below the stronger baseline. The architectural conclusion is confounded with instruction-tuning, since the sequence-to-sequence checkpoints evaluated were never instruction-tuned, and the morphosyntactic task is not distinct from part-of-speech tagging, its gold labels being 96.8% identical. The authors have requested that the paper be retracted.

This paper reports the same model outputs, re-scored. It shares the lexicon and the saved generations with that version and supersedes it; no figure here is taken from it, and the dataset counts, matching rules and baselines are all regenerable from the artifacts listed in Section 8.

Table	Command (in scripts/)	Artifact	Job log
§3.1 counts	build_merged_eval_set.py --dry-run --verify	Combined_POS_ Lexicon.csv	—
§3.1 5,701 set	build_sense_split_eval_set.py --verify	*_sense_ split.json	—
§3.3 agreement	compute_iaa.py	*_Annotated.xlsx	—
§5.1, §5.2	rescore_tasks.py --models \$ROSTER	rescored_ 2026-09-28.json	eval_ 75167.out
§5.3 LLMs	compute_bleu.py	translation_ fidelity_bleu.json	eval_tf_ rerun_76683.out
§5.3 NLLB	probe_nllb_translation.py	nllb_probe_*.json	nllb_ sweep_77617.out
§5.3 few-shot	probe_fewshot_translation.py	fewshot_probe_ gemma-7b.json	fewshot_ probe_77601.out

Table 6: Each reported table, the command that regenerates it, the artifact it reads, and the SLURM job that produced the underlying generations.

Readers comparing the two should treat the earlier numbers as withdrawn rather than as an alternative measurement.

8 Reproducibility

Every number in this paper comes from a committed artifact. `$ROSTER` below is the evaluated roster, `gemma-2b`, `gemma-7b`, `mistral-7b`, `falcon-7b`, `qwen-7b`, `mt5-small`, `mt5-large`, `byt5`; passing it pins the tables to these eight models, since later runs add result files to the same directories.

Raw model outputs for every item are in
`results/evaluation_reports/`
`<task>/<task>_<model>.json`.

9 Limitations

- POS labels are the source dictionary’s, not independently re-annotated. The agreement coefficients in Section 3.3 describe a 500-item subset that is alphabetical and contains no verbs, and whose adjudication is assembled from the two annotator columns rather than decided separately. They are not an estimate for the corpus; a stratified re-annotation covering verbs would be needed for that.
- Morphosyntactic features cover 185 of 5,783 gold rows.
- Lexical alignment gold is 38% positional fallback; Section 5.4 is not a result.
- Entries are dictionary headwords, not running text; performance here does not predict sentence-level performance.
- The model set cannot separate architecture from instruction-tuning. An instruction-tuned seq2seq model (`flan-t5`, `mt0`) would be needed.
- The English-only baseline shows the POS task is substantially solvable without Tigrinya. A harder task would withhold the English gloss.

References

- Ahmed Abdelali et al. LARA-Bench: Benchmarking arabic AI with large language models. *arXiv preprint arXiv:2305.14982*, 2023. doi: 10.48550/arXiv.2305.14982.
- Ife Adebara, Hawau Olamide Toyin, Nahom Tesfu Ghebremichael, AbdelRahim Elmadany, and Muhammad Abdul-Mageed. Where are we? evaluating LLM performance on african languages. *arXiv preprint arXiv:2502.19582*, 2025. doi: 10.48550/arXiv.2502.19582.
- Ebtesam Almazrouei et al. The Falcon series of open language models. *arXiv preprint arXiv:2311.16867*, 2023. doi: 10.48550/arXiv.2311.16867.
- Khalid Alnajjar, Mika Härmäläinen, Jack Rueter, and Niko Partanen. Ve’rdd. narrowing the gap between paper dictionaries, low-resource NLP and community involvement. *arXiv preprint arXiv:2012.02578*, 2020. doi: 10.48550/arXiv.2012.02578.
- Jinze Bai et al. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2023. doi: 10.48550/arXiv.2309.16609.
- Lucas Bandarkar et al. The Belebele benchmark: a parallel reading comprehension dataset in 122 language variants. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024. doi: 10.18653/v1/2024.acl-long.44.
- Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960. doi: 10.1177/001316446002000104.
- Alexis Conneau et al. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020. doi: 10.48550/arXiv.1911.02116.
- Cheikh M. Bamba Dione et al. MasakhaPOS: Part-of-speech tagging for typologically diverse african languages. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, 2023. doi: 10.48550/arXiv.2305.13989.
- Fitsum Gaim and Jong C. Park. Natural language processing for tigrinya: Current state and future directions. *arXiv preprint arXiv:2507.17974*, 2026. doi: 10.48550/arXiv.2507.17974.
- Gemma Team, Google DeepMind. Gemma: Open models based on Gemini research and technology. *arXiv preprint arXiv:2403.08295*, 2024. doi: 10.48550/arXiv.2403.08295.
- Naman Goyal et al. The FLORES-101 evaluation benchmark for low-resource and multilingual machine translation. *Transactions of the Association for Computational Linguistics*, 10: 522–538, 2022. doi: 10.48550/arXiv.2106.03193.
- Suchin Gururangan et al. Annotation artifacts in natural language inference data. In *Proceedings of NAACL-HLT*, 2018. doi: 10.48550/arXiv.1803.02324.
- Amr Hendy et al. How good are GPT models at machine translation? a comprehensive evaluation. *arXiv preprint arXiv:2302.09210*, 2023. doi: 10.48550/arXiv.2302.09210.
- Albert Q. Jiang et al. Mistral 7B. *arXiv preprint arXiv:2310.06825*, 2023. doi: 10.48550/arXiv.2310.06825.
- Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293, 2020. doi: 10.48550/arXiv.2004.09095.

- Katharina Kann, Ophelie Lacroix, and Anders Søgaard. Weakly supervised POS taggers perform poorly on truly low-resource languages. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020. doi: 10.48550/arXiv.2004.13305.
- Abdeljalil Kirouane and Christos Petroucheilos. Thinking in a low-resource language: What SFT builds, what RL fixes, what accuracy cannot see. *arXiv preprint arXiv:2608.17744*, 2026. doi: 10.48550/arXiv.2608.17744.
- Edward Loper and Steven Bird. NLTK: The natural language toolkit. In *Proceedings of the ACL Workshop on Effective Tools and Methodologies for Teaching Natural Language Processing and Computational Linguistics*, 2002. doi: 10.48550/arXiv.cs/0205028.
- R. Thomas McCoy, Ellie Pavlick, and Tal Linzen. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019. doi: 10.48550/arXiv.1902.01007.
- Timothy Niven and Hung-Yu Kao. Probing neural network comprehension of natural language arguments. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019. doi: 10.48550/arXiv.1907.07355.
- NLLB Team et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022. doi: 10.48550/arXiv.2207.04672.
- Alp Öktem, Mirko Plitt, and Grace Tang. Tigrinya neural machine translation with transfer learning for humanitarian response. In *AfricaNLP Workshop*, 2020. doi: 10.48550/arXiv.2003.11523.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318, 2002. doi: 10.3115/1073083.1073135.
- Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. Hypothesis only baselines in natural language inference. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, 2018. doi: 10.48550/arXiv.1805.01042.
- Maja Popović. chrF: character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395, 2015. doi: 10.18653/v1/W15-3049.
- Matt Post. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation*, pages 186–191, 2018. doi: 10.48550/arXiv.1804.08771.
- Anka Reuel et al. BetterBench: Assessing AI benchmarks, uncovering issues, and establishing best practices. *arXiv preprint arXiv:2411.12990*, 2024. doi: 10.48550/arXiv.2411.12990.
- Nathaniel R. Robinson, Perez Ogayo, David R. Mortensen, and Graham Neubig. ChatGPT MT: Competitive for high- (but not low-) resource languages. *arXiv preprint arXiv:2309.07423*, 2023. doi: 10.48550/arXiv.2309.07423.
- Zhi Rui Tam, Cheng-Kuang Wu, Yi-Lin Tsai, Chieh-Yen Lin, Hung-yi Lee, and Yun-Nung Chen. Let me speak freely? a study on the impact of format restrictions on performance of large language models. *arXiv preprint arXiv:2408.02442*, 2024. doi: 10.48550/arXiv.2408.02442.
- Hailay Teklehaymanot, Dren Fazlija, Niloy Ganguly, Gourab K. Patro, and Wolfgang Nejdl. TiGQA: An expert-annotated question-answering dataset in tigrinya. *arXiv preprint arXiv:2404.17194*, 2024. doi: 10.48550/arXiv.2404.17194.

- Hailay Kidu Teklehaymanot, Gebrearegawi Gebremariam, and Wolfgang Nejdl. LLM probe: Evaluating LLMs for low-resource languages. In *Proceedings of the Workshop on Shaping Multilingual, Multimodal AI for the Social Sciences and Humanities (LLMs4SSH) at LREC 2026*, pages 224–234, 2026a. doi: 10.63317/4ezcsk97smvd. URL <https://aclanthology.org/2026.llms4ssh-1.24/>. ACL Anthology 2026.llms4ssh-1.24.
- Hailay Kidu Teklehaymanot, Demeke Desalegn Yadeta, and Wolfgang Nejdl. Expanding the lexicon of ge’ez based african languages: A comparative study of amharic and tigrinya. *arXiv preprint arXiv:2607.15209*, 2026b. doi: 10.48550/arXiv.2607.15209.
- Abrehalei Tela, Abraham Woubie, and Ville Hautamäki. Transferring monolingual model to low-resource language: The case of tigrinya. *arXiv preprint arXiv:2006.07698*, 2020. doi: 10.48550/arXiv.2006.07698.
- Atnafu Lambebo Tonja et al. Natural language processing in ethiopian languages: Current state, challenges, and opportunities. *arXiv preprint arXiv:2303.14406*, 2023. doi: 10.48550/arXiv.2303.14406.
- Union of International Democrats. Tigrinya–english dictionary. Digitised edition, 2015. <https://uidswansea.com/wp-content/uploads/2015/03/tigrinya-english-dictionary.pdf>.
- Kasun Wickramasinghe and Nisansa de Silva. Sinhala-english parallel word dictionary dataset. In *Proceedings of the International Conference on Industrial and Information Systems (ICIIS)*, 2023. doi: 10.1109/ICIIS58898.2023.10253560.
- Thomas Wolf et al. Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP: System Demonstrations*, pages 38–45, 2020. doi: 10.48550/arXiv.1910.03771.
- Ka Wong, Praveen Paritosh, and Lora Aroyo. Cross-replication reliability — an empirical approach to interpreting inter-rater reliability. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, 2021. doi: 10.48550/arXiv.2106.07393.
- Linting Xue et al. mT5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of NAACL-HLT*, 2021. doi: 10.48550/arXiv.2010.11934.
- Linting Xue et al. ByT5: Towards a token-free future with pre-trained byte-to-byte models. *Transactions of the Association for Computational Linguistics*, 10:291–306, 2022. doi: 10.48550/arXiv.2105.13626.
- Tony Z. Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. Calibrate before use: Improving few-shot performance of language models. In *Proceedings of the 38th International Conference on Machine Learning*, pages 12697–12706, 2021. doi: 10.48550/arXiv.2102.09690.
- Wenhao Zhu et al. Multilingual machine translation with large language models: Empirical results and analysis. *arXiv preprint arXiv:2304.04675*, 2023. doi: 10.48550/arXiv.2304.04675.

A Dataset derivation

This appendix records the row-level derivation summarised in Section 3.1. Every count is reproducible from the released code: `build_merged_eval_set.py --dry-run --verify` rebuilds the gold files from the source CSV and compares them against what is released, and `build_sense_split_eval_set.py --verify` does the same for the 5,701-item set.

Source and provenance. The digitised file `Combined_POS_Lexicon.csv` holds 7,234 rows: 3,587 English→Tigrinya entries followed by 3,647 Tigrinya→English entries. It is a flattening of the working annotation workbook, whose eleven part-of-speech sheets hold the same 7,234 rows with the same per-category counts (3,929 noun, 1,784 verb, 993 adjective, 194 adverb, and so on). The flattening lifts the inline gender and number markers that the workbook keeps inside the Tigrinya field — (nf) ዐካሲት, ከሳሲት — out into the label column, which is the origin of the finer categories (nm) noun masculine, (nf) noun feminine and (npl) noun plural; the two files differ on exactly the 247 rows carrying such a marker. The direction swap is present in the workbook as well, in 3,635 of its 7,234 rows (50.2%), so it originates in digitisation rather than in the export.

7,234 counts rows, not independent annotations. Of the 3,646 unique reverse-direction pairs, 1,448 (39.7%) also occur in the forward block.

From 7,234 to 5,783. Merging the two directions deduplicates on the (English, Tigrinya, features) triple rather than on the pair alone, so that distinct senses of one form survive: *estimate* → ግምት is kept as both noun and verb, and *grocer* → በዓል ድካን as both noun and noun, masculine. Expanding comma-separated senses gives 5,864 atomic pairs.

From 5,783 to 5,775. Each task script applies its own pair-level deduplication when loading the gold, dropping 7 duplicate (English, Tigrinya) pairs and 1 row with no English side.

From 5,775 to 5,701. The two tagging tasks were run against a gold file in which each multi-sense entry is represented by its first sense: (*a little*, ንእሽቶይ, ቁሩብ, ውሕድ) was evaluated as (*a little*, ንእሽቶይ). That intermediate file was not originally released, so the task scripts as published would evaluate 5,775 items rather than the 5,701 behind Sections 5.1 and 5.2. We reconstruct it from the released gold and verify the result row-for-row against the saved model outputs: 5,701 items, no row missing, none extra, and no label mismatch. The reconstructed file is released alongside the gold.